

Use database 'SQLITE' in the cloud

Gitpod:cloud-based development environment platform

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What is Gitpod?



Gitpod Classic

- Gitpod is a cloud-based development environment.
- Automatically launches dev containers from Git repositories.
- Offers browser-based VS Code interface, terminal access, and pre-installed tools.

Why Use Gitpod?

- **Zero Setup:** Start coding instantly without installing dependencies.
- **Portability:** Code from any device with a browser.
- **Collaboration:** Share reproducible environments via URLs.
- **Automation:** Prebuild environments using `.gitpod.yml`.

Gitpod Features



Gitpod

Automated, standardized, and secure development environments

 Continue with Google

 Continue with GitHub

 Continue with SSO

Log in to Gitpod Classic

<https://app.gitpod.io/login>

Free

Best for individuals and testing the waters.

- ✓ Self-host in your Cloud, on Linux-hosts or run on your laptop
- ✓ Up-and-running in less than 3 minutes
- ✓ Up to 10 cloud dev environments
- ✓ Gitpod Desktop with unlimited local environments
- ✓ Integrates with AI coding assistants & agents
- ✓ Available in all cloud regions
- ✓ Up to 896 cores and 12TB RAM & GPU support
- ✓ Dev Container support
- ✓ Custom workflow automations
- ✓ Use any editor, incl. VS Code browser

Using the Gitpod Console

- Gitpod provides a built-in terminal at the bottom of the browser IDE.
- You can run Linux commands just like a local shell.
- Common examples:
 - `ls`, `pwd`, `git clone`
 - Run Python scripts: `python3 myscript.py`
 - Compile LaTeX: `pdflatex main.tex`
- Supports package managers like `apt`, `pip`, `npm`.

Tip: You can open multiple terminals or split terminal view for multitasking.

Installing Docker Images in Gitpod

- Gitpod supports Docker via **Docker-in-Docker (dind)**.
- Steps:
 - 1 Open terminal in Gitpod.
 - 2 Start Docker (if needed): `sudo service docker start`
 - 3 Pull an image: `docker pull ubuntu`
 - 4 Run a container: `docker run -it ubuntu bash`
- You can also build images from a Dockerfile:
 - `docker build -t myimage .`

Note: Add Docker support in `.gitpod.yml` and `.gitpod.Dockerfile`.

Gitpod vs Oracle Cloud (Free Tier)

Feature	Gitpod	Oracle Cloud Free Tier
Purpose	Dev Environments (IDE)	Cloud Infrastructure (VMs)
Setup Time	Instant	Manual provisioning
Preinstalled Tools	Yes	No
Browser IDE	Yes	No (SSH/VNC)
Best For	Coding	Hosting/Servers
Free Usage	50 hours/month	2 VMs, always free

What is a Database?

- A **database** is an organized collection of data that can be easily accessed, managed, and updated.
- Data is typically structured in rows and columns (like tables).
- Example: A school might store student info, grades, and attendance in a database.

Types: Relational like MySQL, NoSQL like MongoDB, Graph Database like MemGraph, etc.

What is a DBMS?

- A **Database Management System (DBMS)** is software that allows users to create, read, update, and delete data in a database.
- It ensures:
 - Data integrity
 - Security and access control
 - Concurrent access by multiple users
- **Examples of DBMS:**
 - Relational: MySQL, PostgreSQL, SQLite, Oracle
 - NoSQL: MongoDB, Firebase, Cassandra

Update Anomaly (Without DBMS)

- Suppose our university stores data in a spreadsheet:
 - Student Name, Department, Advisor Email
- If a professor changes their email, we need to update it in every row.
- **Problem:** If one row is missed, data becomes inconsistent.

This is called an Update Anomaly.

Update Anomaly Example: University Spreadsheet

- Suppose, Gandaki University maintains student data in a spreadsheet:

Student data in a Spreadsheet:

Student Name	Department	Advisor Email
Mira	MIT AI	saroj@gu.edu
Ram	MIT AI	kapil@gu.edu
Hari	BIT	saroj@gu.edu
...	...	...
...	...	...
Sita	BIT	saroj@gu.edu
Gita	MIT AI	kapil@gu.edu

Problem: If Saroj Sir changes his email, we must update it in multiple rows.

Risk: Forgetting even one row causes inconsistency.

Solving Update Anomaly with DBMS

- Normalize the data using two separate tables:
 - **Advisors Table** (stores unique advisor info)
 - **Students Table** (references advisor by name or ID)

Advisors Table:

Advisor Name	Email
Saroj	new_saroj@gu.edu
Kapil	kapil@gu.edu

Students Table:

Student Name	Department	Advisor Email
Mira	MIT AI	Saroj
Ram	MIT AI	Kapil
Hari	BIT	Saroj
...	...	...
...	...	...
Sita	BIT	Saroj
Gita	MIT AI	Kapil

Benefit: Just 1 update of Saroj Sir's email in the Advisors table is enough.

Why Use a Database and DBMS?

- Avoids redundancy and inconsistencies.
- Enables data sharing among users.
- Ensures data security and backup.
- Supports data integrity and constraints.
- Efficient query and transaction management.

DBMS solves:

- Update anomalies
- Data duplication
- Unauthorized access

A Simple Example: SQLite

- **SQLite** is a lightweight, serverless relational database.
- Stores the entire database in a single file.
- Great for prototyping, mobile apps, and small applications.

Example SQL Commands:

- `CREATE TABLE students (id INT, name TEXT);`
- `INSERT INTO students VALUES (1, 'Alex');`
- `SELECT * FROM students;`

No installation needed — runs with a single executable.

Installing SQLite and sqlite-web in Gitpod

- Gitpod supports installing packages using apt and pip.
- **Install SQLite:**
 - `sudo apt update`
 - `sudo apt install sqlite3`
- **Install sqlite-web (via pip):**
 - `pip install sqlite-web`
- **Run sqlite-web:**
 - `sqlite-web mydatabase.db`
- Opens a local web interface (port forwarding enabled by Gitpod).

New Workspace

Create a new workspace in the **bdur's Org** organization.



empty

github.com/gitpod-io/empty



VS Code · 1.97.2

Editor · Browser



Standard

Class · Up to 4 cores, 8GB RAM, 30GB storage



Continue (⌘Enter)

Gitpod install sqlite

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

 bash + ▾ □

```
● gitpod /workspace/empty (main) $ sudo apt install sqlite
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libsqlite0 sqlite3
Suggested packages:
  sqlite-doc sqlite3-doc
The following NEW packages will be installed:
  libsqlite0 sqlite sqlite3
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 945 kB of archives.
After this operation, 2,330 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libsqlite0 amd64 2.8.17-15fakesync1build1 [160 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/universe amd64 sqlite amd64 2.8.17-15fakesync1build1 [16.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 sqlite3 amd64 3.37.2-2ubuntu0.4 [769 kB]
Fetched 945 kB in 1s (720 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libsqlite0.
(Reading database ... 36571 files and directories currently installed.)
Preparing to unpack .../libsqlite0_2.8.17-15fakesync1build1_amd64.deb ...
Unpacking libsqlite0 (2.8.17-15fakesync1build1) ...
Selecting previously unselected package sqlite.
Preparing to unpack .../sqlite_2.8.17-15fakesync1build1_amd64.deb ...
Unpacking sqlite (2.8.17-15fakesync1build1) ...
Selecting previously unselected package sqlite3.
Preparing to unpack .../sqlite3_3.37.2-2ubuntu0.4_amd64.deb ...
Unpacking sqlite3 (3.37.2-2ubuntu0.4) ...
Setting up libsqlite0 (2.8.17-15fakesync1build1) ...
Setting up sqlite (2.8.17-15fakesync1build1) ...
Setting up sqlite3 (3.37.2-2ubuntu0.4) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.10) ...
```

Gitpod install sqlite web

- **gitpod /workspace/empty** (main) \$ python
Python 3.12.11 (main, Jun 17 2025, 09:20:38) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> exit()
- **gitpod /workspace/empty** (main) \$ **pip install sqlite-web**
Collecting sqlite-web
 Downloading sqlite-web-0.6.4.tar.gz (807 kB)

807.6/807.6 kB 29.6 MB/s eta 0:00:00

 Preparing metadata (setup.py) ... done
Collecting flask (from sqlite-web)
 Downloading flask-3.1.1-py3-none-any.whl metadata (3.0 kB)

Gitpod install sqlite web

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Preparing metadata (setup.py) ... done
Collecting flask (from sqlite-web)
Downloading flask-3.1.1-py3-none-any.whl metadata (3.0 kB)

Sqlite database

```
○ gitpod /workspace/empty (main) $ sqlite3 testsqlitebd.db
SQLite version 3.37.2 2022-01-06 13:25:41
Enter ".help" for usage hints.
sqlite> CREATE TABLE users (
    id INTEGER PRIMARY KEY ,
    name TEXT NOT NULL,
    age INTEGER
);
sqlite> insert into users values (1,"John",32);
sqlite> insert into users values (1,"Mary",12);
Error: stepping, UNIQUE constraint failed: users.id (19)
sqlite> insert into users values (2,"Mary",12);
sqlite> select * from users;
1|John|32
2|Mary|12
sqlite> █
```

Sqlite database ...

- **gitpod** /workspace/empty (main) \$ sqlite3 testsqlitebd.db
SQLite version 3.37.2 2022-01-06 13:25:41
Enter ".help" for usage hints.
sqlite> insert into users values (3,"Henry John",32);
sqlite> select * from users;
1|John|32
2|Mary|12
3|Henry John|32
sqlite> .quit
- **gitpod** /workspace/empty (main) \$ □

Launch sqlite-web for testsqlite.db

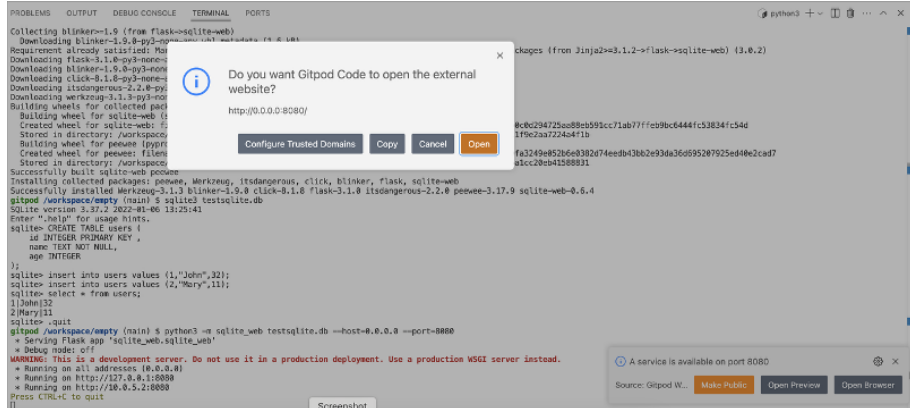
- To launch the SQLite web interface for the database file testsqlite.db, use the following command in the terminal:

Command

```
python3 -m sqlite_web testsqlite.db --host=0.0.0.0  
--port=8080
```

- **Note:** The Python module is written as `sqlite_web` (hyphen '-' replaced by underscore '.').
- This command:
 - Starts a web server bound to all IPs (0.0.0.0).
 - Serves the database GUI on port 8080.
 - Makes the interface publicly accessible.
- **Access:** Open the forwarded URL provided by Gitpod or your terminal output.

Launch sqlite-web for testsqlite.db



```
Collecting blinker==1.9 (from flask->sqlite-web)
  Downloading blinker-1.9.0-py3-none-any.whl (8.3 kB)
Requirement already satisfied: Flask in /workspace/.venv/lib/python3.10/site-packages (from flask->sqlite-web) (3.0.2)
Downloading flask-3.1.0-py3-none-any.whl (101 kB)
Downloading blinker-1.9.0-py3-none-any.whl (8.3 kB)
Downloading click-8.1.8-py3-none-any.whl (109 kB)
Downloading itsdangerous-2.2.0-py3-none-any.whl (26 kB)
Downloading werkzeug-3.1.3-py3-none-any.whl (227 kB)
Building wheels for collected packages:
  Building wheel for sqlite-web (pyproject.toml)
    Created wheel for sqlite-web:
      Stored in directory: /workspace/.venv/lib/python3.10/site-packages
    Building wheel for peewee (pyproject.toml)
    Created wheel for peewee:
      Stored in directory: /workspace/.venv/lib/python3.10/site-packages
Successfully built sqlite-web peewee
Installing collected packages: peewee, Werkzeug, itsdangerous, click, blinker, flask, sqlite-web
Successfully installed Werkzeug-3.1.3 blinker-1.9.0 click-8.1.8 flask-3.1.0 itsdangerous-2.2.0 peewee-3.17.9 sqlite-web-0.6.4
gitpod /workspace/empty (main) $ sqlite3 testsqlite.db
sqlite version 3.37.2 2022-01-06 13:29:41
Enter ".help" for usage hints.
sqlite> CREATE TABLE users (
  id INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  age INTEGER
);
sqlite> insert into users values (1,"John",32);
sqlite> insert into users values (2,"Mary",11);
sqlite> select * from users;
1|John|32
2|Mary|11
sqlite> .quit
gitpod /workspace/empty (main) $ python3 -m sqlite_web testsqlite.db --host=0.0.0.0 --port=8080
* Serving Flask app 'sqlite_web.sqlite_web'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8080
* Running on http://10.0.5.2:8080
Press CTRL+C to quit
```

Do you want Gitpod Code to open the external website?

<http://0.0.0.0:8080/>

[Configure Trusted Domains](#) [Copy](#) [Cancel](#) [Open](#)

A service is available on port 8080

Source: Gitpod W... [Make Public](#) [Open Preview](#) [Open Browser](#)

Create table flight_tab

flight_tab	Create	Query
	Drop	Import
		Export

Figure: Create New table "flight_tab"

Create table flight_tab

Download "flights-export.csv"

Structure

Content

Query

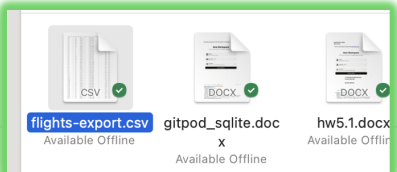
Insert

Import data

Input file

Choose File

flights-export.csv



Supported file types: **JSON** and **CSV**. JSON files must be structured as a list of objects

Create columns for unrecognized keys?

Yes

For instance, if your table has one column *name* and the import file contains objects with

```
SELECT * FROM flight_tab  
LIMIT 10;
```

Query 1. Show some data (limit 10 rows)

```
SELECT Year,  
       SUM(CAST(passenger_count AS INTEGER)) AS Total_Passengers,  
       AVG(CAST(passenger_count AS INTEGER)) AS Average_Passengers  
FROM flight_tab GROUP BY Year;
```

Query 2. Calculate the total and average number of passengers per year

```
SELECT Year,  
       SUM(CAST(passenger_count AS INTEGER)) AS Total_Passengers,  
       AVG(CAST(passenger_count AS INTEGER)) AS Average_Passengers  
FROM flight_tab  
WHERE CAST(Net_Income_Fin_Position_ AS INTEGER) > 100 GROUP BY Year;
```

Query 3. Total and average passengers per year, where Net Income > 100

```
SELECT Year,  
       SUM(CAST(passenger_count AS INTEGER)) AS Total_Passengers,  
       AVG(CAST(passenger_count AS INTEGER)) AS Average_Passengers  
FROM flight_tab  
WHERE CAST(Year AS INTEGER) BETWEEN 2010 AND 2015 GROUP BY Year;
```

Query 4. Total and average passengers per year from 2010 to 2015

Thank You!

References and Resources

- **SQLite Official Website:** <https://sqlite.org>
- **sqlite-web Documentation:**
<https://github.com/sqlite-web/sqlite-web>
- **Gitpod Official Website:** <https://www.gitpod.io>
- **Gitpod Docs (Getting Started):**
<https://www.gitpod.io/docs/introduction/getting-started>
- **Gitpod Examples on GitHub:** <https://github.com/gitpod-io>